KSHITIJ JAIN

919-527-8541 | jkshtj@outlook.com | https://jkshtj.github.io/

PROFESSIONAL SUMMARY

Compiler engineer specializing in MLIR-based toolchains for AI accelerators. Strong background in bridging AI model semantics with specialized hardware features/constraints via principled compiler engineering. Experienced in performance-critical systems, from JIT compilation to high-throughput distributed services.

EXPERIENCE

Compiler Engineer, d-Matrix Inc. *June 2024 - Present*

Developing MLIR-based compiler toolchain for d-Matrix's in-memory compute inference accelerator.

- Implemented d-Matrix hardware-specific compiler micro-optimizations for GEMM and GEMM+SIMD workloads for a number of different dimensions.
- Led LLAMA3 8B model bring-up. Implemented weight externalization enabling compilation on memory-constrained systems (16GB RAM vs. previous 32GB+ requirement). Defined scale-out partitioning scheme for distributed execution across multiple devices.
- Architected padding infrastructure throughout compiler pipeline to handle models with dimensions misaligned to d-Matrix's hardware word size, enabling compilation of a number of popular transformer-based models such as, CLIP, Whisper, GPT2, Llama3 etc.
- Designed the compiler interface for ingesting and allocating hand-written performance kernels, integrating them into primary compilation pipeline while maintaining optimization opportunities across kernel boundaries.

Compiler Engineer, PassiveLogic Inc. *June 2023 - June 2024*

Worked on the Swift compiler, optimizing performance of Differentiable Swift and its runtime infrastructure.

- Implemented middle-end compiler optimizations, including constant-folding, inlining, and most importantly Autodiff closure specialization, achieving 2x performance improvement on PassiveLogic's public Autodiff benchmarks through optimized memory usage.
- Improved performance-critical runtime components supporting Differentiable Swift, including memory manager for efficient tape-based reverse-mode AD.
- Commits can be seen at github.com/apple/swift/commits?author=jkshtj.

Software Development Engineer, AWS *March 2018 - May 2023*

Worked on high-performance systems across AWS RDS and S3.

OPEN SOURCE PROJECTS

- **LLVM JITLink ELF/i386 Backend** (https://blog.llvm.org/posts/2023-03-16-adding-new-llvm-jitlink-target-object-backend/, https://github.com/llvm/llvm-project/commits?author=jkshtj)
- **Microc Compiler** (https://github.com/jkshtj/Microc)

Skills

- **Programming Languages** - C++, Rust, Python, C
- **Compiler Technologies** - MLIR, LLVM, XLA, PyTorch

Education

Purdue University | B.Sc. Computer Engineering | December 2017